# Extreme Heat: spot it, cool it, save a life

Extreme heat is the deadliest type of weather in many countries, yet many heat deaths can be prevented or reduced. The two things that matter most: **cool the body down fast** and **check on people at high risk before they get into trouble.**

CALL EMERGENCY SERVICES NOW

### Heat stroke is a life-threatening emergency

Call your local emergency number (911 in the US/Canada, 112 in much of Europe, 999 in the UK) **immediately** if a person is very hot and shows **confusion, slurred speech, fainting, seizures, or loss of consciousness.** While you wait, start cooling them right away (see below). Do not wait to see if they improve.

## Know the two stages

HEAT EXHAUSTION — ACT NOW

### What you may see

- Heavy sweating
- Cool, pale, clammy skin
- Fast, weak pulse
- Nausea, vomiting
- Muscle cramps
- Tiredness, weakness, dizziness, headache
- Fainting

### What to do

- Move to a cool place (shade or air conditioning)
- Loosen clothing
- Sip cool water
- Apply cool, wet cloths or take a cool bath

**Get medical help** if the person is vomiting, symptoms get worse, or symptoms last longer than 1 hour.

HEAT STROKE — EMERGENCY

### What you may see

- Body temperature of 103°F (39.4°C) or higher
- Hot, red skin that is dry *or* damp
- Fast, strong pulse
- Confusion, slurred speech
- Throbbing headache, dizziness
- Losing consciousness

### What to do

- **Call emergency services immediately**
- Move the person somewhere cool
- Lower their temperature with cool cloths or a cool bath
- **Do not give them anything to drink** if they are confused or unconscious

## Who is most at risk

Heat does not affect everyone equally. Check on these people often during a heat wave:

- Adults aged 65 and older
- Babies and young children
- People with chronic conditions such as heart disease, high blood pressure, diabetes, kidney disease, or mental illness
- People taking certain medications (some affect the body's ability to stay cool — ask a pharmacist if unsure)
- Outdoor workers and athletes
- People who live alone or without air conditioning
- People who are pregnant

PREVENTION

### Stay ahead of the heat

- **Drink water** regularly — don't wait until you feel thirsty. (If a doctor limits your fluids, ask them how much to drink in heat.)
- **Stay cool** — spend time in air conditioning. If your home is hot, go to a public cooling center, library, or shopping mall.
- **Avoid the hottest hours** — limit strenuous activity to early morning or evening.
- **Dress light** — wear lightweight, loose, light-colored clothing.
- **Take it easy** — rest often in the shade and pace outdoor work.

NEVER LEAVE ANYONE IN A PARKED CAR

The temperature inside a parked car can rise about 20°F (11°C) in just 10 minutes, even with a window cracked. **Never leave children, older adults, or pets in a parked vehicle** — not even for a minute.

BUDDY-CHECK TOOLS

### Turn this guide into a check-in network

- **Read the Evidence-Based Cooling Guide** for practical cooling steps, fan caveats, hydration, and home heat workarounds.
- **Print and share the Heat Safety Flyer** — a high-contrast, easy-to-read one-page community guide with local fill-in fields.
- Print a heat buddy-check sheet for neighbors, relatives, tenants, coworkers, or community members.
- Use the 60-minute setup plan to assign buddies, gather local resources, and follow up.
- Copy SMS and phone scripts for friendly, privacy-preserving check-ins.

## Be a good neighbor

Most people who die in heat waves are found alone. A simple check-in can save lives:

- Pick 2–3 at-risk neighbors, friends, or relatives and agree to check on each other during heat warnings.
- Call or visit in person twice a day during extreme heat.
- Make sure they have water, a way to stay cool, and a working phone.
- Know the route to the nearest cooling center and offer to go together.

## Sources


- U.S. Centers for Disease Control and Prevention (CDC) — *Heat & Health* and warning signs of heat-related illness.
- U.S. National Weather Service (NWS) — *Heat Safety Tips and Resources*.
- World Health Organization (WHO) — *Heat and health* fact sheet.
- Ready.gov (U.S. Dept. of Homeland Security) — *Extreme Heat*.


Guidance summarized from the public-health sources above. Temperature and symptom thresholds follow CDC/NWS guidance; emergency-number examples are general.

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# Wildfire Smoke: breathe safe, help others

Wildfire smoke carries tiny particles that enter your lungs and bloodstream. Much smoke exposure can be reduced. The two things that matter most: **stay aware of air quality** and **reduce smoke exposure indoors and out.**

MEDICAL EMERGENCY

**Smoke can trigger life-threatening reactions**

Call your local emergency number **immediately** if someone has chest pain, severe shortness of breath, confusion, or lips/fingernails turning blue. People with asthma or COPD should follow their emergency action plan and seek care if rescue inhalers do not help.

## Air Quality Index — know the colors

The AQI tells you how dangerous the air is right now. Check an official local air-quality source when available, such as **AirNow.gov** in the U.S., your national/regional environment agency, or a trusted local weather alert service.

0–50 GREEN

**Good**

Air quality is fine. Normal activities are safe for most people.

51–100 YELLOW

**Moderate**

People unusually sensitive to smoke, including children, older adults, and people with asthma/COPD or heart disease, should consider reducing prolonged outdoor exertion.

101–150 ORANGE

**Unhealthy for sensitive groups**

Sensitive groups should avoid prolonged or heavy outdoor exertion; everyone else should consider reducing it, especially if symptoms appear.

151–200 RED

**Unhealthy**

Everyone should avoid prolonged exertion outdoors. Sensitive groups should stay inside.

**201–300 PURPLE**

### Very unhealthy

Everyone should reduce activity, stay indoors in cleaner air if possible, and keep windows closed unless indoor heat creates a bigger immediate danger.

**301+ MAROON**

### Hazardous

Emergency conditions. Avoid outdoor exertion. If you must go out and can wear one safely, use a well-fitting N95, KN95, or P100 respirator.

## Who is most at risk?

- Children and teenagers (lungs still developing)
- Adults 65 and older
- Pregnant people
- People with asthma, COPD, or other lung diseases
- People with heart disease or diabetes
- Outdoor workers
- People experiencing homelessness or without reliable shelter

## Protect yourself and others

1. **Check the AQI every morning** during wildfire season. Set alerts on AirNow or your weather app.
2. **Stay indoors in cleaner air** with windows and doors closed when smoke is unhealthy, unless indoor heat or another hazard makes that unsafe.
3. **Create a clean-air room.** Choose an interior room with few windows. Run an air purifier continuously. Avoid candles, frying, vacuuming, or gas stoves without ventilation.
4. **Wear a well-fitting N95, KN95, or P100 respirator** if you must go outside in unhealthy smoke and can wear one safely. Cloth masks do not reliably block PM2.5; respirators may not fit infants/young children and may be difficult for some people with breathing conditions.
5. **Consider a Corsi-Rosenthal box.** This DIY air cleaner uses a box fan and MERV-13 or better filters and can substantially reduce indoor particle levels when built and used correctly. Follow current build and electrical-safety guidance; use newer, safety-certified fans where possible; and do not block exits or create trip hazards.
6. **Check on vulnerable neighbors daily.** Offer masks, a ride to a clean-air shelter, or help running a filter.
7. **Reduce or cancel outdoor events** when AQI is over 150.

## For community organizers

ACTION CHECKLIST

- Identify or coordinate **clean-air spaces** with HEPA filtration, safe DIY air cleaners, or other verified filtration.
- **Distribute well-fitting respirators** to outdoor workers and unhoused neighbors, with fit/use caveats and local worker-safety guidance.
- Share AQI alerts via text, social media, and local networks.
- Reschedule outdoor community events when AQI > 150.
- Partner with local libraries, community centers, and faith groups to extend shelter hours.

## When to seek medical help

- Difficulty breathing or shortness of breath
- Persistent cough, especially with phlegm or blood
- Chest pain or tightness
- Wheezing that does not improve with medication
- Dizziness or confusion
- Palpitations or irregular heartbeat

## Key resources

- **AirNow.gov** — Real-time AQI and fire maps (U.S.)
- **Fire and Smoke Map** — fire.airnow.gov
- **EPA Wildfire Smoke Guide** — epa.gov/pm-pollution/wildfire-smoke-guide-factsheets
- **Corsi-Rosenthal Box build guides** — cleanaircrew.org
- **Emergency** — 911 (U.S./Canada), 112 (Europe), 999 (UK)

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**Sources include:**

- AirNow — wildfire smoke and air quality information.
- U.S. Environmental Protection Agency (EPA) — wildfire smoke and health guidance.
- U.S. Centers for Disease Control and Prevention (CDC) — wildfire smoke and health.
- U.S. National Weather Service — wildfire safety.
- Published research and public guidance on DIY air filtration; verify device-building instructions against current safety guidance before use.

This guide is for educational purposes and does not replace professional medical advice. In an emergency, call your local emergency number.

# Opioid overdose: recognize it, reverse it

An opioid overdose slows or stops a person's breathing. **Naloxone** is a medicine that can quickly reverse it. It only works on opioids, it is not addictive, and giving it to someone who is *not* overdosing is very unlikely to harm them — so when in doubt, use it and call for help.

CALL EMERGENCY SERVICES FIRST

**Always call for help**

If you think someone is overdosing, call your local emergency number (911 in the US/Canada, 112 in much of Europe, 999 in the UK) **right away**, then give naloxone. Naloxone can wear off before the opioids do, so professional help is still needed even if the person wakes up.

## Know the signs of an overdose

WARNING SIGNS

- Will not wake up or respond to your voice or touch
- Breathing is very slow, shallow, or has stopped
- Gurgling, gasping, or snoring/choking sounds
- Lips, fingertips, or skin turning blue, gray, or pale
- Very small, "pinpoint" pupils
- Limp body; cold or clammy skin

To check responsiveness, call their name, tap or shake their shoulder if safe, and follow local first-aid or dispatcher instructions. If they do not respond and are not breathing normally, treat it as an overdose emergency.

## What to do — step by step

1. **Call emergency services.** Report that someone is unresponsive and not breathing normally, and give your location.
2. **Give naloxone.** For nasal spray (e.g. Narcan): tilt the head back, insert the nozzle into one nostril until your fingers touch the bottom of their nose, and press the plunger firmly all the way in.
3. **Support their breathing.** Follow dispatcher instructions. If trained, give rescue breaths or CPR as appropriate. If not trained, keep their airway clear and stay with them until help arrives.
4. **Wait 2–3 minutes.** If there is no improvement (still not breathing or not awake), give a **second dose** in the other nostril.
5. **Put them in the recovery position.** Once breathing, lay them on their side so they don't choke if they vomit.

6. **Stay with them** until emergency help arrives. Naloxone can wear off in 30–90 minutes and the overdose can return.

GOOD TO KNOW

- **Naloxone is safe to use in an emergency.** It only reverses opioids and is very unlikely to harm someone who has not taken opioids. Do not delay when overdose is possible.
- **It is widely available.** In the U.S., naloxone nasal spray is approved for over-the-counter sale and is free or low-cost through many pharmacies, health departments, and harm-reduction programs. Availability varies by country — ask a local pharmacist or health service.
- **Withdrawal may follow.** Naloxone can cause sudden, uncomfortable opioid withdrawal (sweating, nausea, agitation). Reassure the person, keep them from using more opioids, and keep monitoring until emergency help arrives.
- **Many overdoses involve fentanyl,** a very strong opioid. A second or additional dose of naloxone may be needed if breathing does not improve after 2–3 minutes — follow product directions and dispatcher guidance.

GOOD SAMARITAN PROTECTIONS

Many places have "Good Samaritan" laws that protect people who call for help during an overdose from certain drug-possession charges. Protections vary by location — but **calling for help is always the right thing to do.** A life comes first.

## Be prepared before an emergency

- Carry naloxone if you or someone you know uses opioids (including prescription painkillers).
- Ask a pharmacist or local health department how to get it and how to use your specific product.
- Tell the people around you where your naloxone is kept.
- Encourage people not to use alone; if they do, ask them to tell someone or use a "never use alone" hotline where available.

## Sources


- U.S. Substance Abuse and Mental Health Services Administration (SAMHSA) — overdose prevention and naloxone guidance.
- U.S. Centers for Disease Control and Prevention (CDC) — about naloxone and overdose prevention.
- U.S. National Institute on Drug Abuse (NIDA) — *Naloxone DrugFacts*.
- U.S. Food and Drug Administration (FDA) — naloxone drug facts and over-the-counter naloxone information.


Steps reflect SAMHSA/CDC overdose-response guidance. Always follow the instructions printed on your specific naloxone product, and your local emergency services' direction.

# Dehydration & ORS: rehydrate early, know danger signs

Dehydration can become dangerous fast, especially for babies, young children, older adults, and people who are ill. **Oral rehydration solution (ORS)** replaces water and salts more effectively than plain water when dehydration risk is high.

**SEEK URGENT MEDICAL HELP NOW**

Get urgent help for lethargy, confusion, unconsciousness, seizures, inability to drink or keep fluids down, repeated vomiting, blood in stool, severe belly pain, signs of severe dehydration, symptoms in a very young infant, or any person who is getting worse despite rehydration. Use local emergency services when needed.

## Use packaged ORS if available

Use commercially prepared ORS packets when you can. Mix exactly as directed on the packet, using safe water and the full amount of water listed. Too little water can make the solution too salty; too much water can make it less effective.

## If no packet is available: emergency home mix

**MEASURE CAREFULLY**

Only use a home mix when packaged ORS is not available and you can measure carefully.

- 1 liter safe water
- 6 level teaspoons sugar
- 1/2 level teaspoon salt

Stir until dissolved. The drink should taste no saltier than tears. If it tastes very salty, throw it away and mix again. Discard after 24 hours, or sooner if contaminated.

## How to give fluids

- Give small, frequent sips. If vomiting occurs, wait 5–10 minutes and try again slowly.
- For infants and young children, use a spoon, cup, or syringe in small amounts often.
- Continue breastfeeding and normal age-appropriate feeding when possible.
- Do not give fluids by mouth to someone who is unconscious, very confused, unable to swallow safely, or having a seizure.
- Avoid alcohol. Avoid very sugary drinks as the main rehydration fluid during diarrhea; they can worsen diarrhea for some people.

## For children: also give zinc

RECOMMENDED BY WHO & UNICEF

For a child with diarrhea, the WHO and UNICEF recommend giving **zinc every day for 10–14 days**, alongside ORS. Zinc shortens the illness, makes it less severe, and helps prevent diarrhea in the following months. The usual amount is **20 mg per day** for children, and **10 mg per day for infants under 6 months**. Use a zinc product made for children and follow the package or a health worker's instructions, and keep giving it for the full course even after the diarrhea stops.

## Watch for dehydration

CONCERNING SIGNS

- Very thirsty, dry mouth, little or no urination
- Dizziness, weakness, fast heartbeat
- Sunken eyes, no tears in a child, unusual sleepiness
- Persistent diarrhea or vomiting

EMERGENCY SIGNS

- Very drowsy, confused, limp, or unconscious
- Unable to drink or breastfeed
- Repeated vomiting or signs getting worse
- Blood in stool, severe pain, or seizure

## For caregivers and volunteers

- Ask age, symptoms, whether the person can drink, urine frequency, vomiting/diarrhea frequency, and whether blood is present.
- Prioritize urgent care for babies, older adults, pregnant people, people with chronic illness, and anyone getting worse.
- Do not diagnose the cause of diarrhea or vomiting. Help with safe fluids and escalation.
- Use safe water. If water safety is uncertain, follow local guidance for boiling, treating, or using sealed water.

## Prevent the next illness

- **Wash hands with soap** after using the toilet or cleaning a child, and before preparing food or eating.
- **Use safe drinking water,** stored covered and clean; boil or treat water when its safety is uncertain.
- **Use a toilet or latrine** and dispose of stool, including children's, safely.
- **Vaccinate:** the rotavirus vaccine protects babies against a common, severe cause of diarrhea — ask your health service.

## Sources


- World Health Organization (WHO) — diarrhoeal disease and oral rehydration guidance.
- UNICEF — diarrhoeal disease and oral rehydration resources.

- WHO/UNICEF — joint statement on clinical management of acute diarrhoea, including ORS and zinc.
- U.S. Centers for Disease Control and Prevention (CDC) — diarrheal illness prevention and dehydration-related guidance.
- MedlinePlus / U.S. National Library of Medicine — dehydration information.

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This guide is general information, not medical advice, and is not a substitute for professional care. Local guidance may differ, especially for infants, cholera outbreaks, severe malnutrition, kidney/heart disease, or other medical conditions. When in doubt, seek medical help.

# Choking: clear the airway fast

Choking happens when something blocks the airway and stops air from getting to the lungs. If the person can still cough or speak, the blockage is partial — encourage them to keep coughing. If they **cannot breathe, speak, or cough,** the airway is badly blocked and you must act right away.

CALL EMERGENCY SERVICES

**Severe choking is an emergency**

Call your local emergency number (911 in the US/Canada, 112 in much of Europe, 999 in the UK), or have someone else call, if a person is choking and **cannot breathe, speak, cry, or cough**, is clutching their throat, is going silent or blue, or becomes unconscious. If you are alone with them, give first aid first and call as soon as you can — put the phone on speaker.

## First, tell mild from severe

MILD — LET THEM COUGH

- Can still cough forcefully, speak, or breathe
- Coughing is the most effective way to clear it

**Encourage them to keep coughing.** Do not slap their back or do thrusts yet; stay with them and be ready to act if it gets worse.

SEVERE — ACT NOW

- Cannot breathe, speak, cry, or make sounds
- Silent or very weak cough; clutching the throat
- Lips or face turning blue; looking panicked

**Begin first aid immediately** and get emergency help.

## Adult or child (over 1 year)

1. **Give up to 5 back blows.** Stand to the side and slightly behind. Support their chest with one hand and lean them well forward. With the heel of your other hand, give up to five firm blows between the shoulder blades. Check after each one to see if the blockage has cleared.
2. **Give up to 5 abdominal thrusts.** Stand behind them and put your arms around their waist. Make a fist just above the navel and below the ribcage; grasp it with your other hand and pull sharply inward and upward, up to five times.
3. **Keep alternating** 5 back blows and 5 abdominal thrusts until the object comes out, the person can breathe, or they become unconscious.

**FOR A PREGNANT PERSON OR A VERY LARGE PERSON**

Do not give abdominal thrusts. Instead give **chest thrusts**: place your fist on the center of the breastbone and pull straight back, sharp and firm, alternating with back blows.

## Baby (under 1 year)

**NEVER GIVE A BABY ABDOMINAL THRUSTS**

Use back blows and chest thrusts only. Support the head at all times.

1. **5 back blows.** Lay the baby face-down along your forearm with the head lower than the body, supporting the jaw and head. Give up to five firm blows between the shoulder blades with the heel of your hand.
2. **5 chest thrusts.** Turn the baby face-up along your other forearm, head still lower than the body. Place two fingers on the center of the chest (lower half of the breastbone) and give up to five quick, firm thrusts.
3. **Keep alternating** 5 back blows and 5 chest thrusts. Call emergency services if the baby does not start breathing, and continue until help arrives or the baby becomes unresponsive.

## If the person becomes unconscious

**START CPR**

1. Lower them carefully to the floor and **call emergency services now** (speaker phone) if not already done.
2. Begin **CPR**: chest compressions, and rescue breaths if you are trained and willing. If you are not trained, follow the dispatcher's instructions; they may guide hands-only CPR until help arrives.
3. Each time you open the airway to give breaths, **look in the mouth** and remove any object you can clearly see. Do not sweep blindly with your finger, which can push it deeper.
4. Continue until the person recovers or emergency help takes over.

**AFTER A SEVERE CHOKING EPISODE**

Anyone who needed back blows or thrusts — or who has a lasting cough, trouble swallowing, or a feeling that something is still stuck — should be checked by a medical professional, because thrusts can occasionally cause internal injury and small fragments can remain.

## Prevent choking, especially in young children

- Cut food for young children into small pieces; encourage everyone to eat slowly and not talk or laugh with a full mouth.

- For children under about 4, avoid or modify high-risk foods: whole grapes and cherry tomatoes (cut them up), nuts, popcorn, hard or sticky candy, chunks of meat or hot dog, and raw hard vegetables.
- Keep small objects — coins, button batteries, marbles, small toy parts, deflated balloons — out of reach of babies and toddlers.
- Supervise young children while they eat, and keep them seated.

## Sources


- American Red Cross — conscious choking first aid guidance (back blows and abdominal thrusts; infant back blows and chest thrusts).
- American Heart Association (AHA) — first aid guidelines for choking and CPR.
- Resuscitation Council UK — *Choking* first-aid guidance.
- U.S. CDC — choking and suffocation prevention and national child-safety guidance.


Steps reflect widely taught Red Cross / Resuscitation Council first-aid sequences for lay responders. Different training bodies vary in detail — follow your local first-aid training and emergency dispatcher's instructions.

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# Severe bleeding: stop the bleed

Most bleeding you will see is minor and stops on its own or with a little pressure. But bleeding that is **spurting, pooling, soaking through clothing or bandages, or coming from a deep wound to an arm or leg** can be life-threatening. The single most important thing you can do is press hard on the wound and not let up.

CALL EMERGENCY SERVICES

**Life-threatening bleeding is an emergency**

Call your local emergency number (for example 911 in the US/Canada, 112 in much of Europe, 999 in the UK), or have someone else call, for spurting blood, pooling blood, bleeding that soaks through clothing/bandages, an amputation, or any heavy bleeding you are worried about. If you are alone, it is reasonable to **start pressure first** and call as soon as you can — put the phone on speaker and follow dispatcher instructions. Get yourself safe first if there is traffic, fire, weapons, or other danger.

## Three actions that stop bleeding

If you have gloves or any barrier (a plastic bag can help), use them to protect yourself from blood. If no barrier is available and the bleeding is life-threatening, do the best you safely can and wash thoroughly afterward. Find where the blood is coming from — you may need to move or cut away clothing to see the wound clearly.

1. **Press hard with both hands.** Place a clean cloth, gauze, or your protected hands directly on the wound and push down as hard as you can, right on the spot the blood is coming from. Keep steady, firm pressure — do not lift up to check unless a dispatcher or trained responder tells you to. Many serious bleeds slow or stop when pressure is applied firmly enough and held long enough.
2. **If it keeps bleeding, pack the wound.** For a deep wound in a place you can pack — an arm, leg, armpit, or groin (*not* the chest, abdomen, neck, or head) — push clean gauze or cloth firmly all the way into the wound, filling it completely, then press down hard over the top with both hands and keep pressing.
3. **For a limb that won't stop, use a tourniquet.** If heavy bleeding from an arm or leg does not stop with pressure and packing, or the injury is an amputation/near-amputation, apply a tourniquet (see below) if one is available or you are trained to improvise. For life-threatening limb bleeding, tourniquets can be the right choice — follow local training and dispatcher instructions.

## How to use a tourniquet

FOR ARMS AND LEGS ONLY

Use a manufactured tourniquet if you have one. Improvised tourniquets are harder to make effective and can injure tissue; use one only when life-threatening limb bleeding cannot be controlled another way and no manufactured tourniquet is available. Avoid thin cord or rope because it can cut the skin.

1. **Place it 5–7 cm (2–3 inches) above the wound**, between the wound and the heart. Do not place it directly over a joint (knee or elbow) — go above the joint if needed. Never put it on the wound itself.
2. **Pull it tight and tighten until the bleeding stops.** A tourniquet that is tight enough to work will hurt — that is expected. If one is not enough to stop the bleeding, apply a second one just above the first.
3. **Note the time** you put it on (write it on the tourniquet, tape, or a visible note if you can) and tell the medics. **Do not loosen or remove it** — leave that to medical professionals.

## While you wait for help

KEEP PRESSURE AND WATCH FOR SHOCK

- **Keep pressing** the whole time, even once bleeding seems controlled, until help takes over.
- Help the person **lie down** and keep them warm with a coat or blanket; heavy blood loss makes people cold and can cause shock.
- Reassure them and keep talking to them. Watch their breathing and alertness.
- **Do not remove an object stuck in a wound** (knife, glass, metal). Leave it in place and press firmly *around* it — pulling it out can make bleeding much worse.

## Minor cuts and scrapes

- Wash your hands, then rinse the wound under clean running water and gently clean around it.
- Apply gentle pressure with a clean cloth for a few minutes until it stops, then cover with a clean dressing or bandage.
- Watch for signs of infection over the next days — spreading redness, warmth, swelling, pus, or fever — and seek care if they appear. Ask a health worker about tetanus protection for dirty or deep wounds and animal bites.

## Sources


- American College of Surgeons — *Stop the Bleed* (apply pressure, pack the wound, apply a tourniquet).
- American Red Cross — first aid guidance for severe bleeding and wound care.
- World Health Organization — basic emergency care approach to the acutely ill and injured.

- U.S. Centers for Disease Control and Prevention (CDC) — tetanus and wound-care related guidance.

Steps reflect widely taught *Stop the Bleed* / first-aid sequences for lay responders. Training bodies vary in detail, especially around wound packing and improvised tourniquets — follow your local first-aid training and emergency dispatcher's instructions.

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# Hands-only CPR for sudden cardiac arrest

Sudden cardiac arrest is when the heart suddenly stops pumping blood. The person collapses, stops responding, and stops breathing normally (or only gasps). Without blood flow to the brain, every minute counts. Starting CPR right away can improve the chance of survival.

CALL EMERGENCY SERVICES FIRST

**Collapsed and not breathing normally = emergency**

Call your local emergency number (911 in the US/Canada, 112 in much of Europe, 999 in the UK) immediately, or tell a specific person to call and to find an AED. Put the phone on speaker so the dispatcher can guide you. Do not wait to be sure — if someone is unresponsive and not breathing normally, start CPR.

## Three quick checks

1. 1 **Check for a response.** Tap their shoulders firmly and shout, “Are you OK?” If they do not respond, they need help now.
2. 2 **Check breathing.** Look at the chest for no more than about 10 seconds. If they are not breathing, or are only gasping, struggling, or making occasional noisy breaths, treat it as cardiac arrest.
3. 3 **Call and get an AED.** Call emergency services (speaker phone) and send someone for the nearest automated external defibrillator (AED). If you are alone with a phone, call first, then start CPR.

Gasping — slow, noisy, irregular breaths in the first minutes — is **not** normal breathing. It is a sign of cardiac arrest. Start CPR.

## How to do hands-only CPR (adults & teens)

**PUSH HARD, PUSH FAST, DON'T STOP**

1. **Kneel beside them** on a firm, flat surface. Place the heel of one hand on the **centre of the chest** (on the breastbone, between the nipples). Put your other hand on top and interlock your fingers.
2. **Push down hard and fast.** Keep your arms straight and your shoulders directly over your hands, and press straight down at least **5 cm (2 inches)** deep — about one-third the depth of the chest — but not more than about 6 cm (2.4 inches).
3. **Let the chest come all the way back up** between pushes (full recoil) without lifting your hands off. This lets the heart refill.
4. **Keep a fast, steady rate of 100–120 pushes per minute** — about two per second. The beat of many well-known upbeat songs works as a guide.
5. **Do not stop** except to use an AED or to swap with someone. Switch with another bystander about every 2 minutes if you can, to keep pushes strong — good compressions are tiring.

For an untrained rescuer helping an adult or teen, **hands-only CPR** — continuous chest compressions with no rescue breaths — is recommended and effective. If you are trained and willing, give 30 compressions then 2 rescue breaths, but never let giving breaths delay or interrupt good chest compressions for long.

## Use an AED as soon as one arrives

**AEDS ARE MADE FOR UNTRAINED BYSTANDERS**

An AED is a small device, often found in airports, schools, gyms, offices, and shopping centres. **Turn it on and follow the spoken instructions** — it tells you exactly what to do. Stick the pads on the person's bare chest as shown in the pictures on the pads. Keep doing chest compressions while the pads go on. When the AED says “analysing,” make sure no one is touching the person. If it advises a shock, make sure everyone is clear, then press the button. Right after any shock or any “no shock” message, **resume compressions immediately** and keep following the AED.

## When to stop

- Emergency responders or another trained person takes over.
- The person clearly wakes up, moves, opens their eyes, and starts to breathe normally. If so, roll them onto their side (recovery position) and keep watching their breathing.
- An AED instructs you to stop while it analyses or delivers a shock (resume immediately afterward).
- You are too exhausted to continue and no one can take over.

## Children, babies, drowning, and overdose

BREATHS MATTER MORE HERE

In babies and children, and when collapse is caused by drowning, choking, or a drug overdose, lack of oxygen is usually the cause, so **rescue breaths are especially valuable**. If you are trained, give breaths with compressions (commonly 30 compressions to 2 breaths; trained child/infant responders often use 15:2 with two rescuers). For a baby, use two fingers in the centre of the chest and push about 4 cm (1.5 inches) deep. If you are not trained, the dispatcher can guide you — and doing chest compressions alone is far better than doing nothing.

## Good to know

- **Do not wait because you are worried about being perfect.** The biggest mistake is doing nothing. A person in cardiac arrest needs immediate help.
- **Cracked ribs can happen** and are not a reason to stop if the person is in cardiac arrest — effective compressions can save a life.
- **Don't check for a pulse** if you are untrained — it wastes time and is easy to get wrong. Go by responsiveness and breathing.
- A heart attack (a blocked artery, often with chest pain) is different from cardiac arrest, but a heart attack can lead to cardiac arrest — so if someone with chest pain collapses and stops breathing normally, start CPR.
- **Learn CPR in person** if you can. Short, low-cost or free classes are widely available and build the confidence to act.

## Sources


- American Heart Association (AHA) — Hands-Only CPR and AHA CPR & first-aid guidelines (compression rate 100–120/min, depth at least 2 in / 5 cm, full recoil, minimise interruptions).
- American Red Cross — CPR/AED training and resources for adults, children, and infants.
- Resuscitation Council UK — resuscitation and basic life support guidance library (adult and paediatric).
- European Resuscitation Council (ERC) — CPR guidelines.


Steps reflect widely taught basic-life-support sequences for lay responders. Training bodies differ in small details — follow your local first-aid training and your emergency dispatcher's instructions.

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# Stroke: spot it fast

Most strokes are caused by a clot blocking blood flow to the brain; some are caused by bleeding in the brain. You cannot tell which from the outside — and you do not need to. What matters is recognising a stroke and getting urgent medical care, because stroke treatments are highly time-sensitive.

CALL EMERGENCY SERVICES NOW

**Stroke is always an emergency — call immediately**

If you see **any one** of the FAST signs below, call your local emergency number (911 in the US/Canada, 112 in much of Europe, 999 in the UK) right away. Do not wait to see if it passes. Use emergency medical transport if available; if a dispatcher gives different local instructions, follow them. Note the time the signs first started.

## Spot a stroke with FAST

1. **F — Face.** Ask the person to smile. Does one side of the face droop, or is it numb? Is the smile uneven?
2. **A — Arms.** Ask them to raise both arms. Does one arm drift downward, or are they unable to lift one?
3. **S — Speech.** Ask them to repeat a simple sentence. Is their speech slurred, strange, or hard to understand? Can they not speak or not understand you?
4. **T — Time.** If you see any of these signs, it is **time to call emergency services immediately**. Note when the signs first appeared — this guides treatment.

## BE-FAST: two more sudden signs

ALSO CALL RIGHT AWAY IF YOU SEE

- **B — Balance.** Sudden loss of balance or coordination, sudden dizziness, or trouble walking.
- **E — Eyes.** Sudden trouble seeing in one or both eyes, or sudden double or blurred vision.

Other sudden warning signs include sudden numbness or weakness (especially on one side of the body), sudden confusion, and a sudden severe headache with no known cause. The key word is **sudden**.

## What to do while you wait for help

**STAY CALM AND KEEP THEM SAFE**

- **Note the time** the signs first appeared, or when the person was last seen well. This helps decide which treatments are possible — tell emergency responders and the hospital team.
- **Do not give food, drink, or any medicine** — including aspirin. A stroke can make swallowing unsafe, and the right treatment depends on the type of stroke, which only the hospital can determine.
- **Stay with them and keep them comfortable.** Help them sit or lie down with the head and shoulders slightly raised, and loosen tight clothing.
- **Note any medicines they take** (especially blood thinners) and any known conditions, for the hospital team.
- If they become unconscious but are breathing normally, roll them onto their side (recovery position) and keep watching their breathing. If they stop breathing normally, start CPR.

**EVEN IF THE SIGNS GO AWAY, STILL CALL**

If stroke signs appear and then disappear within minutes, it may have been a **transient ischaemic attack (TIA, or "mini-stroke")**. This is a serious warning that a major stroke may follow soon. It is still an emergency — call for help and get medical assessment right away, even if the person feels completely fine.

## Why speed matters

- **"Time is brain."** Brain injury can worsen minute by minute during a stroke, so every minute of delay matters.
- Clot-busting medicines and clot-removal procedures can reduce disability for some people, but they only work within limited time windows after the signs begin.
- Emergency medical transport is usually safer than driving when it is available: responders can start care, choose a hospital equipped for stroke, and alert the team to be ready.

## Lower your risk

- Keep blood pressure controlled — high blood pressure is the biggest risk factor for stroke.
- Don't smoke; keep alcohol within low-risk limits; stay physically active; and ask a clinician about weight, diet, and other risk factors if relevant.
- Manage diabetes and cholesterol, and follow treatment for an irregular heartbeat (atrial fibrillation), which raises stroke risk.

## Sources


- American Stroke Association (AHA/ASA) — stroke warning signs and F.A.S.T.
- U.S. CDC — stroke signs and symptoms and what to do.
- Stroke Association (UK) — symptoms of stroke and the FAST test (stroke.org.uk).
- World Health Organization (WHO) — cardiovascular disease and stroke fact sheets.

FAST and BE-FAST are widely taught public-awareness tools. Some strokes do not fit these signs — trust any sudden, unexplained neurological change and call for help. Follow your local emergency dispatcher's instructions.

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# Burns and scalds: first aid

A burn is damage to the skin from heat, hot liquids (a "scald"), electricity, chemicals, or radiation such as the sun. What you do in the first minutes matters. The single most useful action for most burns is simple: cool it with running water, and do not put creams, oils, or home remedies on it.

CALL EMERGENCY SERVICES NOW

**Get emergency help for a serious burn**

Call your local emergency number (911 in the US/Canada, 112 in much of Europe, 999 in the UK) for a burn that is large (bigger than the person's hand), deep, or on the face, hands, feet, genitals, or over a joint; any burn that goes all the way around a limb; any electrical or chemical burn; any burn with trouble breathing, coughing, a hoarse voice, or soot around the nose or mouth (the airway may be burned); and any serious burn in a baby, young child, pregnant woman, or older or frail person. **While you wait for help, start cooling the burn.**

## What to do, step by step

1. **Stop the burning and stay safe.** Move the person away from the heat source. If clothing is on fire, **stop, drop, and roll**, or smother the flames with a blanket. Switch off the power before touching someone with an electrical burn. For a chemical burn, brush off any dry chemical and remove contaminated clothing, protecting your own skin.
2. **Cool the burn with running water for 20 minutes.** Hold the burned area under cool or lukewarm *running* water as soon as possible. Keep cooling for a full 20 minutes — it still helps even up to about 3 hours after the burn. Do **not** use ice, iced water, or very cold water, which can deepen the injury. If there is no running water, use any clean, cool water you can pour over it.
3. **Remove jewellery and tight or soaked clothing** from near the burn before it starts to swell — but do not pull off anything that is stuck to the skin.
4. **Keep the rest of the person warm.** Cooling a large burn can quickly chill the body, especially in children and older people. Cover the parts you are not cooling with a blanket or coat, and avoid cooling so much that they start to shiver.
5. **Cover the burn loosely** once it is cooled. A layer of clean kitchen cling film (plastic wrap) laid over the burn is ideal — lay it on lengthwise, do not wrap it tightly around a limb. A clean, non-fluffy cloth or a clean plastic bag (good for a hand or foot) also works. This eases pain and keeps the wound clean.

**NEVER PUT THESE ON A BURN**

Do **not** apply butter, oil, lard, toothpaste, egg, soy sauce, ash, soil, cow dung, herbs, ice, or "burn creams" and ointments. These do not help, can trap heat, raise the risk of infection, and make it harder for medical staff to see and treat the burn. **Do not burst blisters** — they protect the skin underneath. Cool water and a clean loose cover are what a burn needs.

## Chemical and electrical burns

### Chemical burns

- Protect yourself first — wear gloves or use a barrier if you can.
- Brush off any **dry** powder, then rinse the area with lots of cool running water for at least 20 minutes.
- Remove clothing and jewellery the chemical has touched while rinsing.
- If a chemical is in the eye, rinse the eye gently with clean water for at least 20 minutes and get medical help.
- If you know the chemical, tell the medical team or poison centre.

### Electrical burns

- **Do not touch the person** until you are sure the power is switched off — you can be electrocuted too.
- Electrical burns can be far deeper than they look on the surface and can affect the heart.
- Always seek medical care, even if the skin damage seems small.
- If the person is unresponsive and not breathing normally, call for help and start CPR.

## Smaller burns you can treat at home

A small, shallow burn — red and sore, perhaps with small blisters, no bigger than the person's hand and not on the face, hands, feet, genitals, or a joint — can usually be cared for at home: cool it, cover it loosely, and keep it clean. Simple pain relief such as paracetamol/acetaminophen or ibuprofen can help (follow the label and any local guidance). **Get medical advice if** the burn becomes more painful, swollen, smelly, or oozing, or the person develops a fever — these can be signs of infection. Make sure tetanus protection is up to date for any burn that breaks the skin.

## Babies, children, and others at higher risk

**LOWER THE THRESHOLD TO GET HELP**

Babies and young children, older or frail people, and pregnant women are more easily harmed by burns. Get medical advice for any burn that blisters in a baby or child, and don't wait if you are unsure. Children also chill quickly when a burn is cooled, so keep the rest of the body warm.

## Preventing burns and scalds

- Keep hot drinks, pots, and kettles out of children's reach; turn pot handles inward; use the back rings of the stove.

- Set water heaters to a safe temperature (about 50 °C / 120 °F) and test bath water before a child gets in.
- Keep cooking fires and stoves stable and away from children; keep flammable clothing away from open flames.
- Fit smoke alarms and check them; keep matches, lighters, and chemicals locked away.
- Protect skin from the sun and avoid sunburn.

## Sources


- World Health Organization — Burns fact sheet (global burden, prevention, and first aid: cool with running water, do not apply harmful substances).
- American Red Cross — first-aid training and resources (burn first aid and when to seek care).
- American Heart Association — first-aid guidelines (cooling burns; thermal, chemical, and electrical injuries).
- NHS — Burns and scalds (cool 20 minutes, cover with cling film, do not burst blisters, when to get help).


Guidance reflects widely taught burn first aid for lay responders. Details vary between training bodies and regions — follow your local first-aid training and emergency services.

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# Seizure first aid

Most seizures end by themselves and do not need emergency treatment. Your job is to protect the person, prevent injury, and gently care for them afterwards — and to recognise the few situations that do need an ambulance.

CALL EMERGENCY SERVICES NOW IF...

**When a seizure is an emergency**

- The seizure (shaking/convulsing) lasts **longer than 5 minutes**.
- Another seizure starts soon after the first, or they do not regain awareness between seizures.
- The person is **hurt, has trouble breathing, or does not wake up** after the shaking stops.
- The seizure happens **in water**.
- It is the person's **first-ever seizure**, or you do not know their history.
- The person is **pregnant**, has **diabetes**, or another serious health condition.

Call your local emergency number (911 in the US/Canada, 112 in much of Europe, 999 in the UK). When in doubt, call.

## What to do during a convulsive seizure

1. **Stay calm and stay with them.** Note the time the seizure starts so you know how long it lasts.
2. **Protect them from injury.** Move away hard or sharp objects. Cushion the head with something soft (a folded jacket). Take off glasses and loosen anything tight around the neck.
3. **Do not restrain them** and **do not put anything in their mouth.** Let the movements happen. Holding them down or putting objects (or your fingers) in the mouth can injure them or you — and you cannot stop the seizure that way.
4. **When the shaking stops, roll them onto their side** (recovery position). This keeps the airway clear and lets saliva drain. Check they are breathing.
5. **Stay until they are fully alert.** Reassure them calmly — they may be confused, sleepy, or upset for a while. Do not give food, drink, or medicine until they are fully awake and able to swallow.

COMMON MYTHS — DO NOT DO THESE

- **Never put anything in the mouth** — a spoon, wallet, fingers, or food. A person **cannot swallow their tongue.** Objects can break teeth, block the airway, or be bitten.
- **Do not hold the person down** or try to stop their movements.
- **Do not try to "wake" them** by slapping, shouting, or throwing water.
- Do not give anything to eat or drink until they are fully recovered.

## Not all seizures involve shaking

Some seizures cause staring, confusion, repeated movements (like lip-smacking or fumbling), wandering, or simply being unaware for a short time. The person may not respond normally. **Do not grab or restrain them.** Gently guide them away from danger (roads, stairs, water), speak calmly and quietly, stay with them, and wait for them to come round. They may not remember it afterwards.

## Fever-related seizures in young children

USUALLY BRIEF AND NOT HARMFUL

Young children sometimes have a seizure when they have a high fever (a "febrile seizure"). It is frightening to watch but is usually short and does not cause lasting harm. Give the same first aid: keep them safe, do not restrain or put anything in the mouth, and lay them on their side. Call for emergency help if it lasts more than 5 minutes, if it is their first one, if they have trouble breathing, or if the child seems very unwell or does not recover quickly.

## After the seizure

- Stay calm and kind. Tiredness, confusion, headache, or embarrassment afterwards are normal.
- Give them space and privacy — don't crowd them.
- Note what you saw to tell them or the medical team: how long it lasted, what the body did, and any injury. This information is genuinely helpful.
- If they have a seizure-action plan or rescue medicine and a trained person is present, follow it.

## Sources


- CDC — Seizure first aid (stay, safe, side; what not to do).
- Epilepsy Foundation — Seizure first aid and safety.
- World Health Organization — Epilepsy fact sheet (global burden; first aid; do not restrain or place objects in the mouth).
- NHS — Epilepsy and what to do when someone has a seizure.
- American Red Cross — first-aid training and resources.


Guidance reflects widely taught seizure first aid for lay responders. Details vary between training bodies and regions — follow your local first-aid training and emergency services.

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